



High Temperature Superconductor for Magnet Coils

About this Book

This compilation of experimental data provides “to the best of our knowledge” information on high temperature superconductor tape product which Faraday Factory produced in amount of more than 7,500 km by now. The data are regularly updated to represent the current performance of this popular product.

All data are collected either by employees of Faraday Factory or by others with the use of HTS tape made by Faraday Factory with a respective reference to the source.

The information in this Book is “as is” and may contain occasional errors. Although we pay a great attention to prove the data in this Book, no warranty is given for the accuracy. We suggest readers to use it as a guide and check information independently.

The actual data of the HTS tape, which Faraday Factory supplies, may differ from those presented in this Book. Please refer to order/delivery documents.

In case you have any inquiries about any of data in this Book, or you wish to suggest edits or amendments, or want to receive up-to-date version of this Book, please contact us anytime at info@faradaygroup.com

Basic Specifications

Superconducting Properties

Critical Current

Recommended Lift Factors

Electric Properties

Room-temperature resistivity

Mechanical properties

Delamination Strength

Double Bending Diameter

Joints

Tape Architecture

Customization options available.
If your request differ from the specifications below,
please contact us at sales@faradaygroup.com

Parameter	Value
Tape Length	100 ... 700 m
Tape Width	4.0 ± 0.2 mm
Average I_c @77K, s.f.	100...200A
Strength	> 500 MPa
Critical deformation	≥ 0.4 %
Critical double-bend Ø	≤ 10 mm
Substrate alloy	Hastelloy C-276
Substrate thickness	35 ... 45 µm
Silver stabilization	2 µm HTS side, 1 µm back
Copper stabilization	5 µm each side

Superconducting Properties

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Critical Current

vs Field Angle

77K

20K

vs Field Strength

0-8T - wide Temp range

4K

20K

30K

65K

Recommended Lift Factors

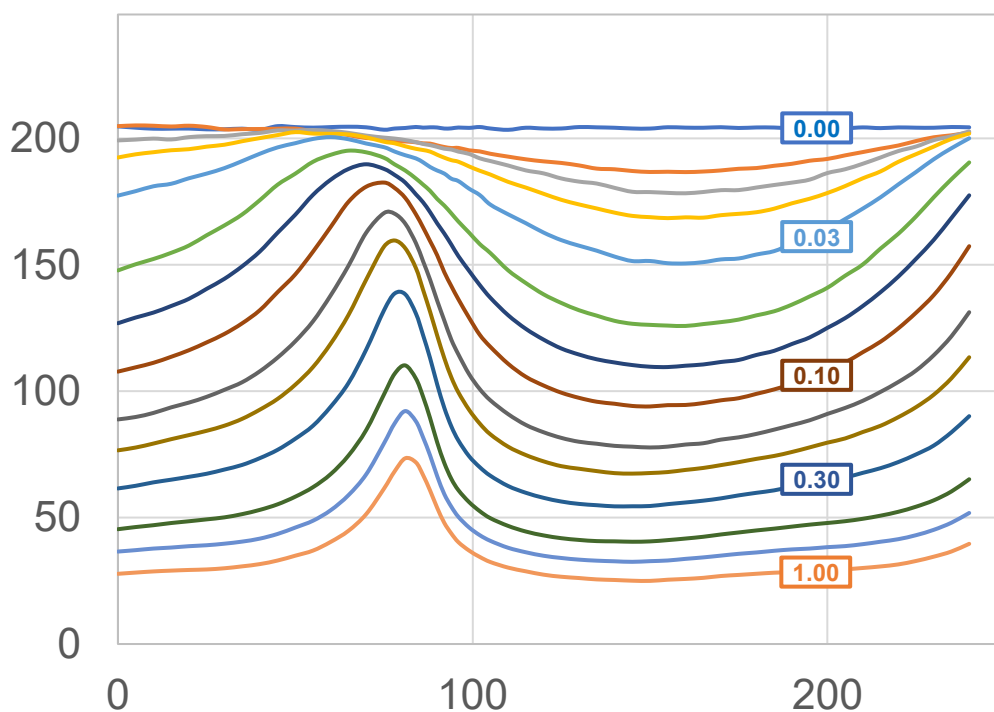
Critical Current

Angular dependence @77K

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I_c @77K
A/4 mm

These are the values measured for a sample with a statistically average performance



Field Angle, deg ($B//c = 0$ and 180 deg)

0.00 – 0.03 T

0.05 – 0.20 T

0.3 – 1.0 T

Instrument: [SuperCurrent, HTS-110](#)

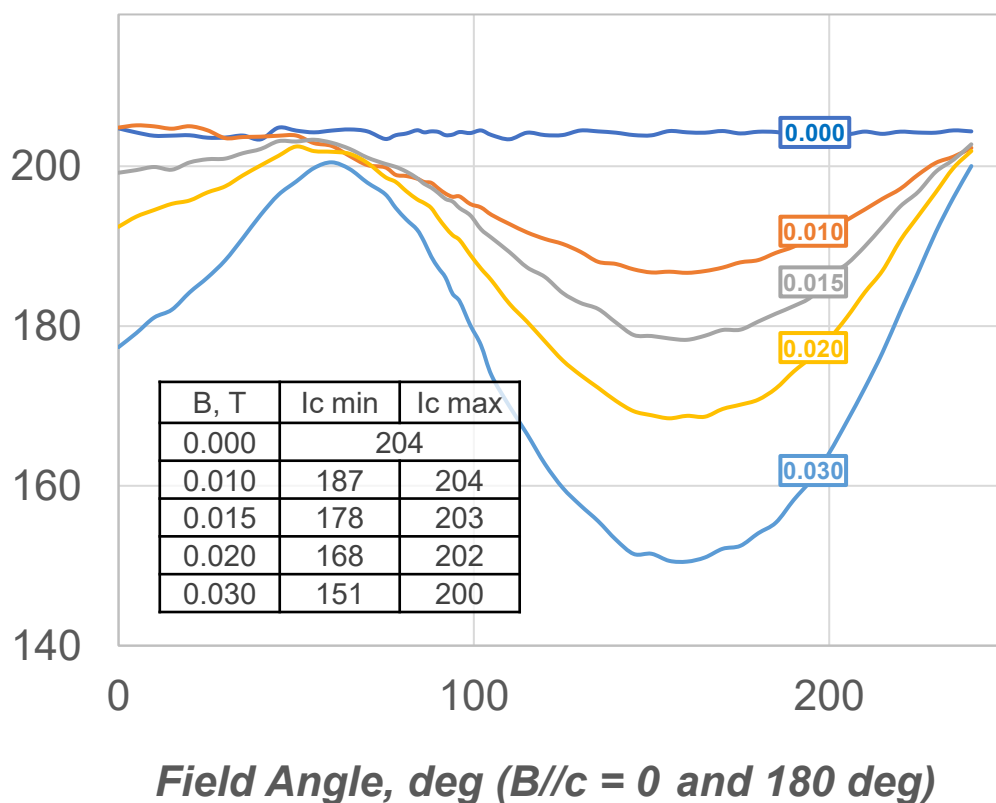
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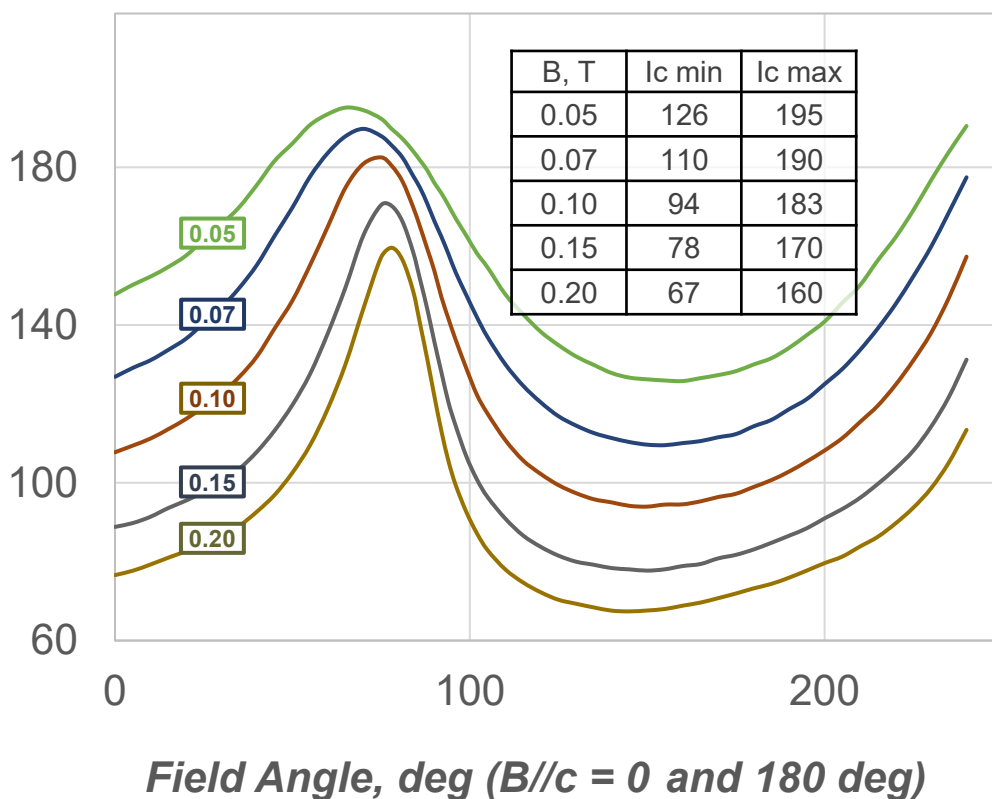
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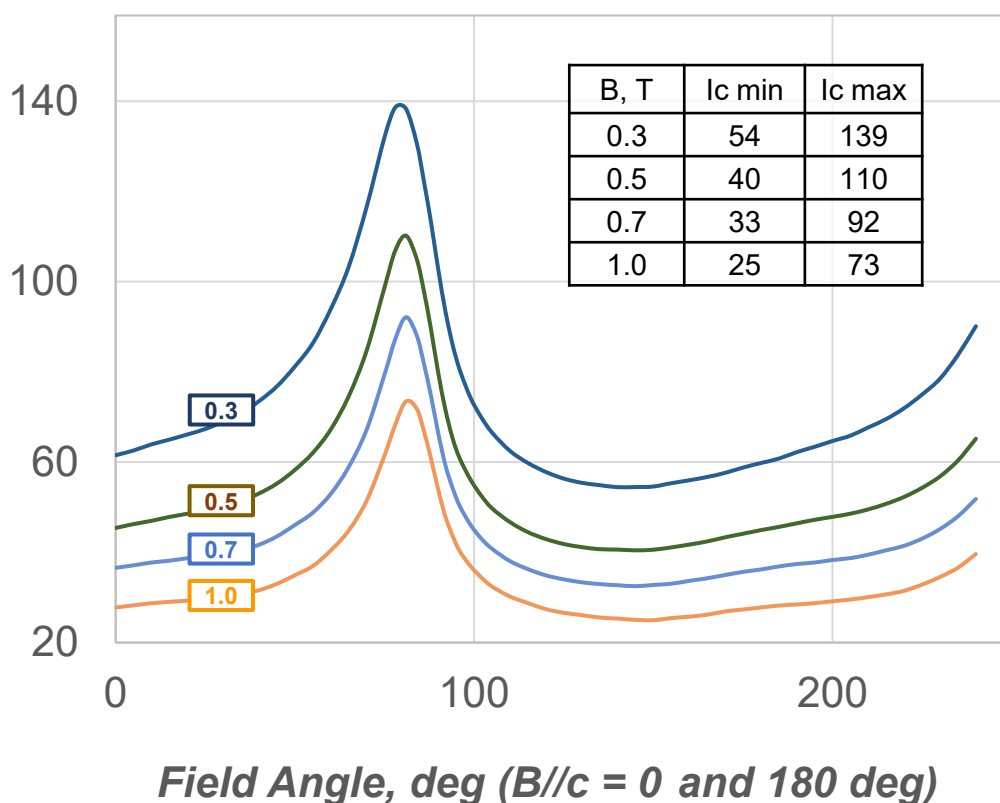
Critical Current

Angular dependence @77K

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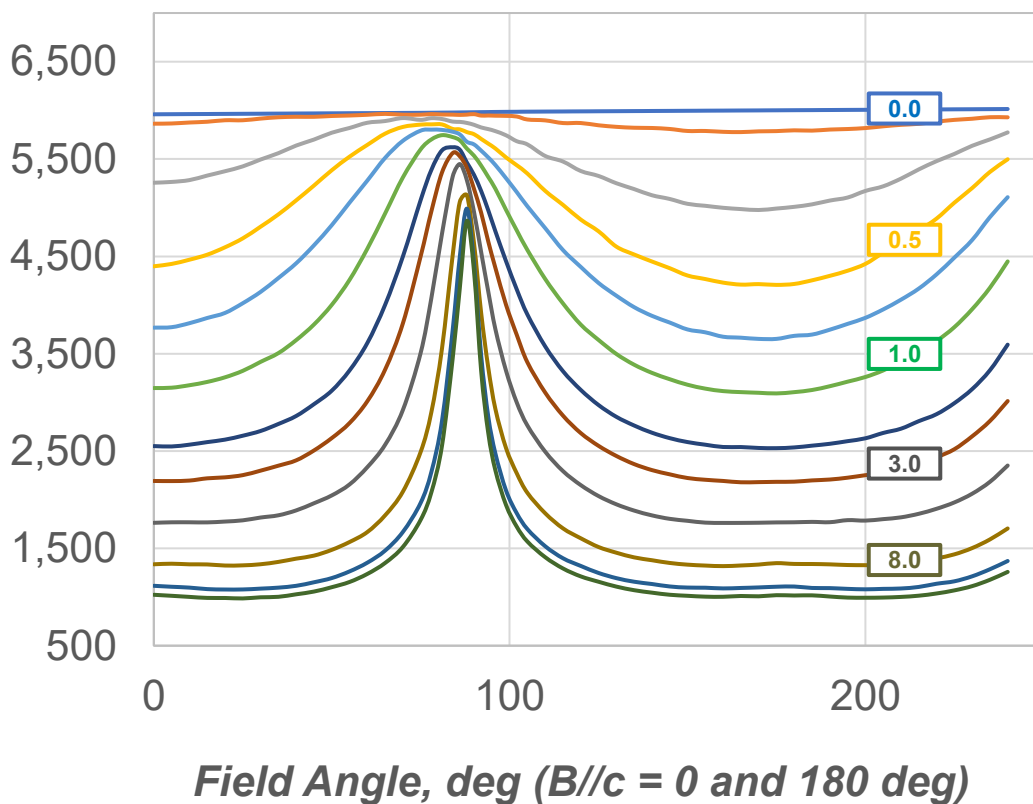
Critical Current

Angular dependence @20K

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I_c @20K
A/4 mm

These are the values measured for a sample with a statistically average performance



0.0 – 0.7 T

1 – 3 T

5 – 8 T

Instrument: [SuperCurrent, HTS-110](#)

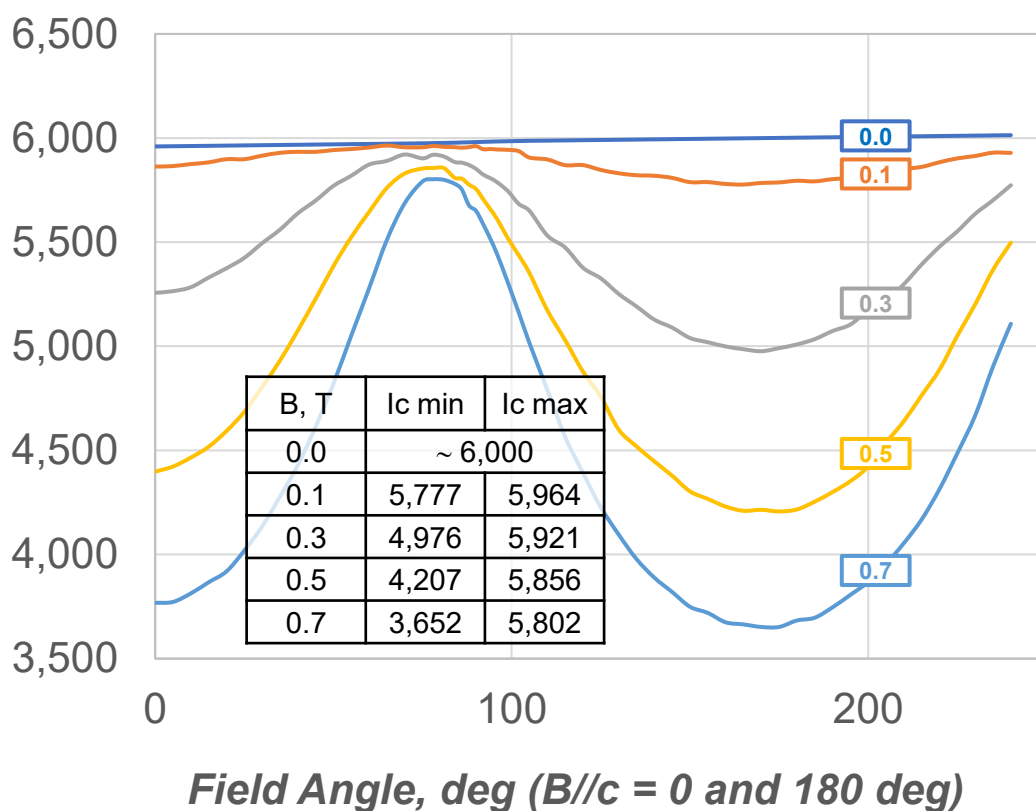
Critical Current

Angular dependence @20K

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I_c @20K
A/4 mm

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0.0 – 0.7 T

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5 – 8 T

Instrument: [SuperCurrent, HTS-110](#)

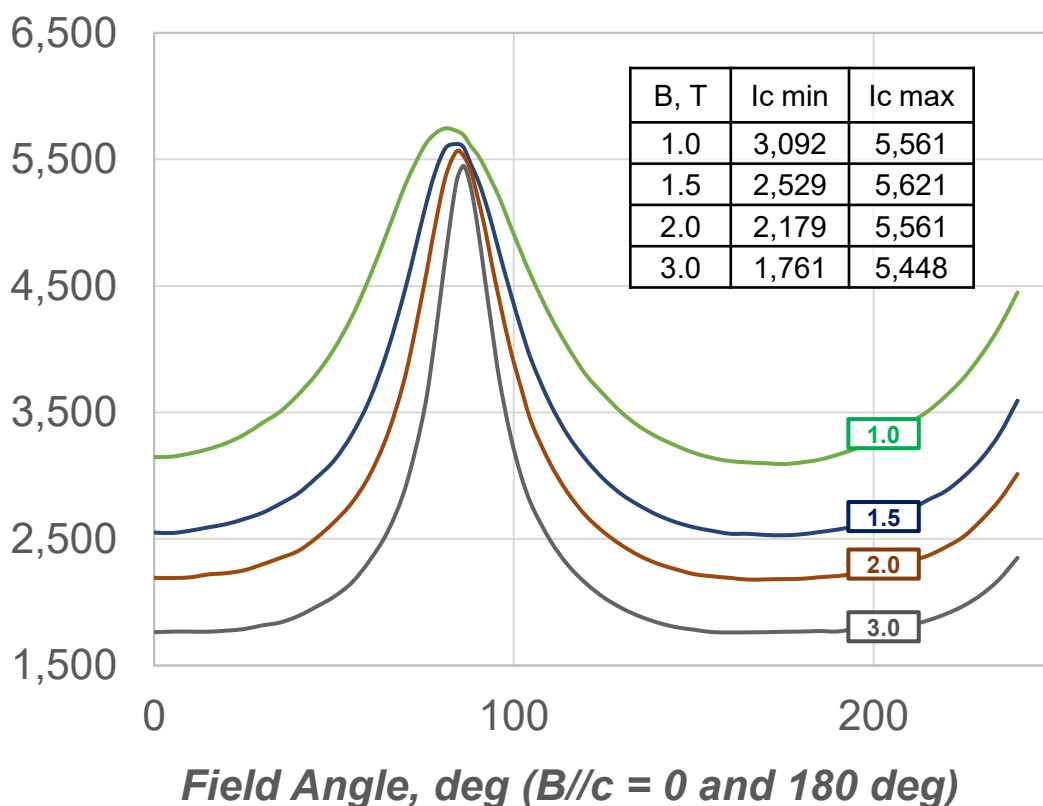
Critical Current

Angular dependence @20K

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1 – 3 T

5 – 8 T

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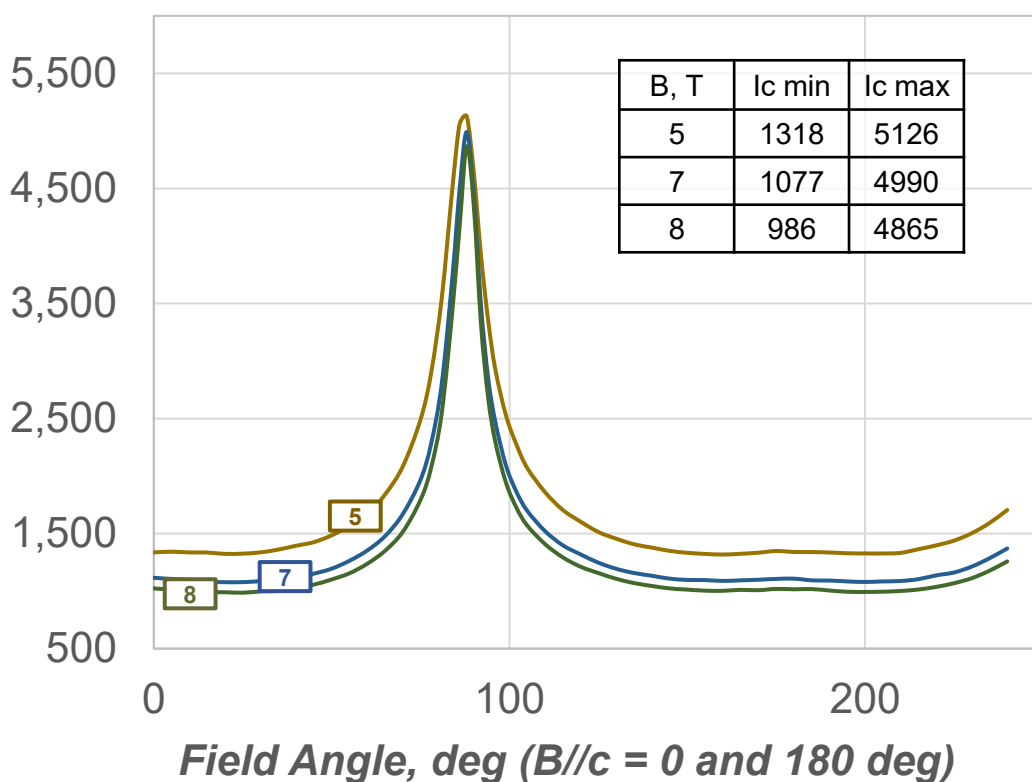
Critical Current

Angular dependence @20K

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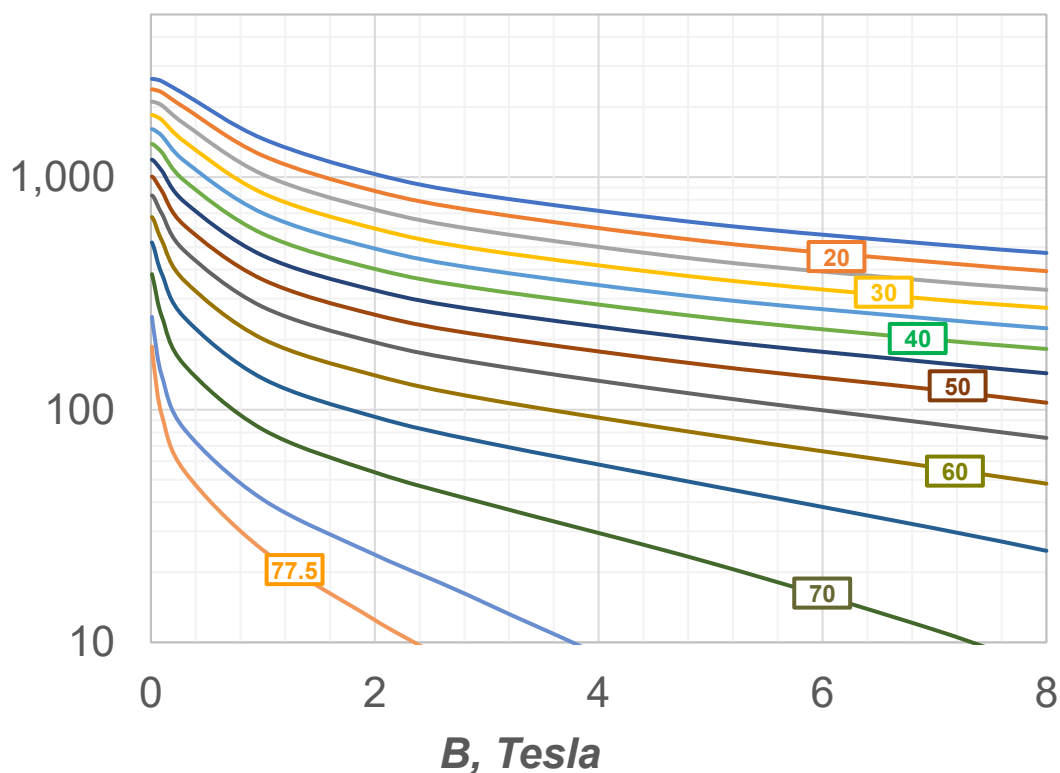
Critical Current

Field dependence 0 ... 8T

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**True min I_c ,
A/4 mm**

These are the values measured for a sample with a statistically average performance



15 – 30 K

35 – 55 K

60 – 77.5 K

Instrument: [SuperCurrent, HTS-110](#)

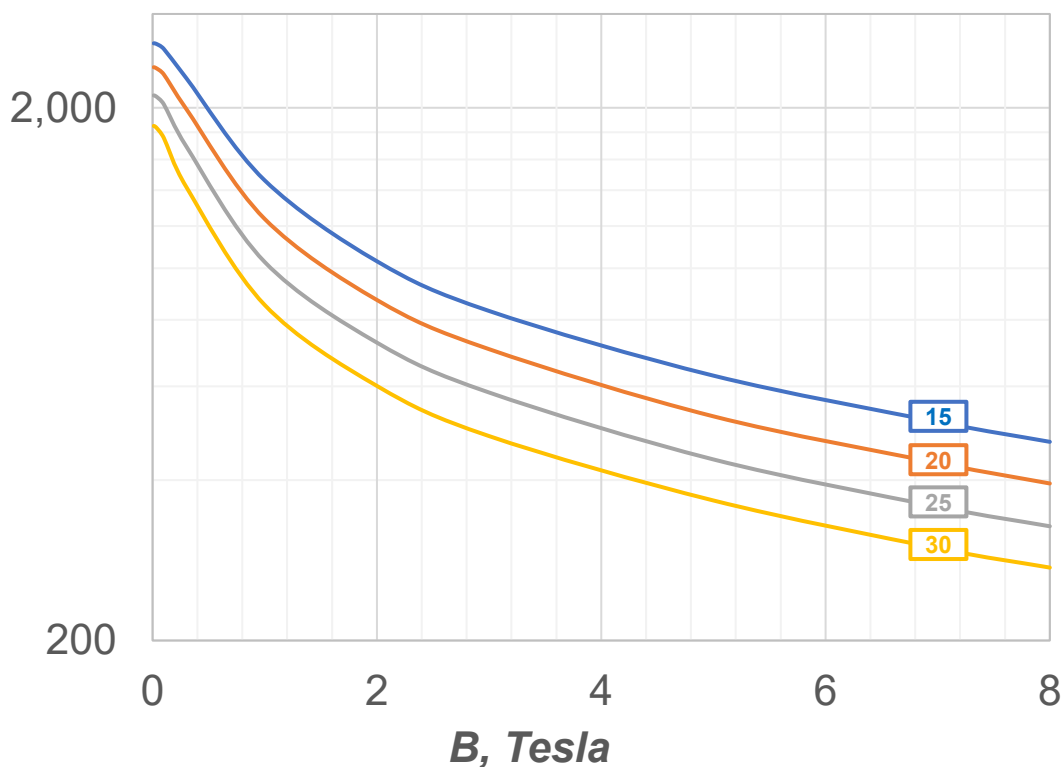
Critical Current

Field dependence 0 ... 8T

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60 – 77.5 K

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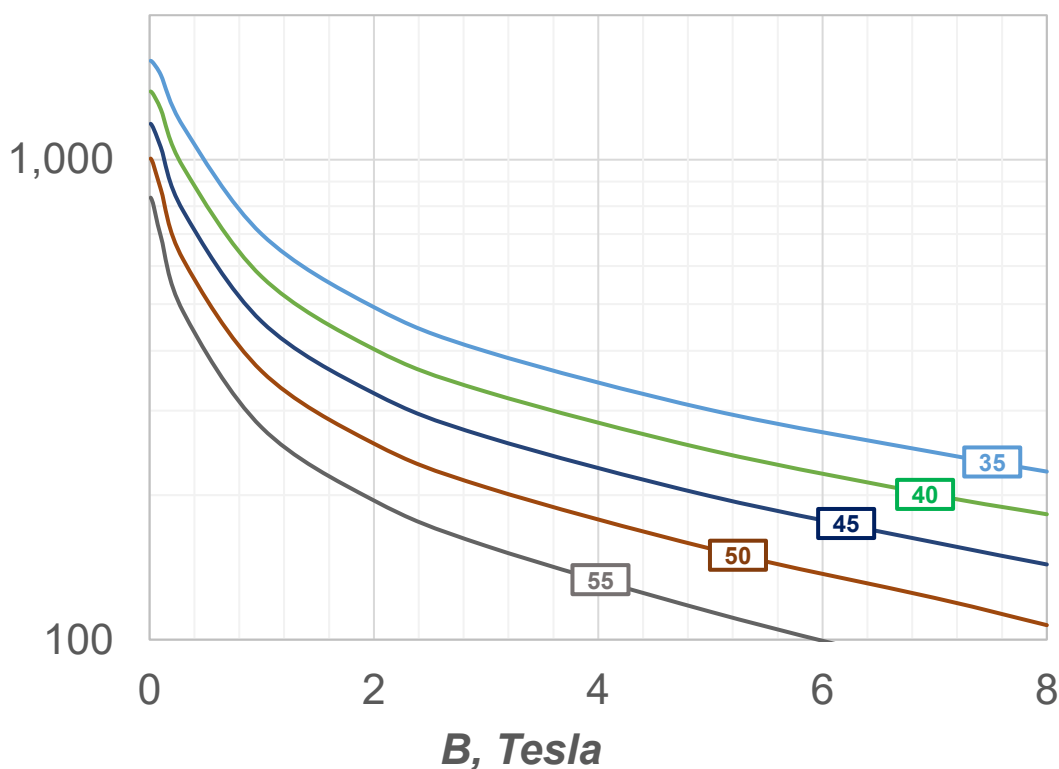
Critical Current

Field dependence 0 ... 8T

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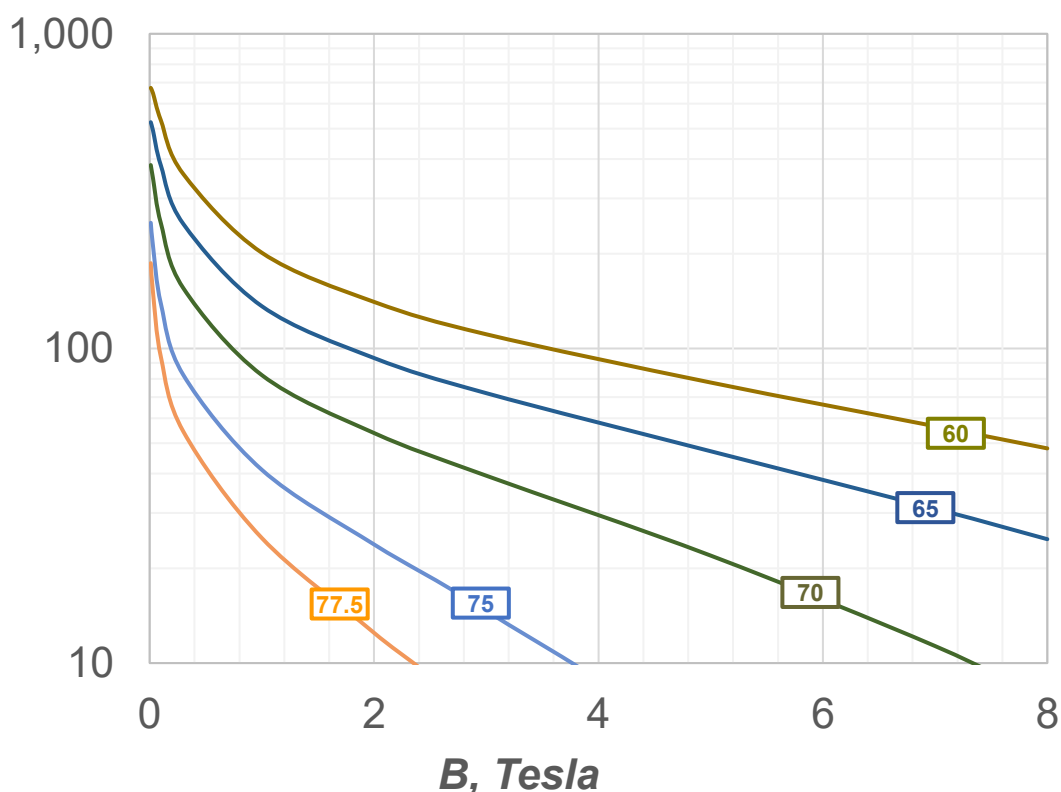
Critical Current

Field dependence 0 ... 8T

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**True min I_c ,
A/4 mm**

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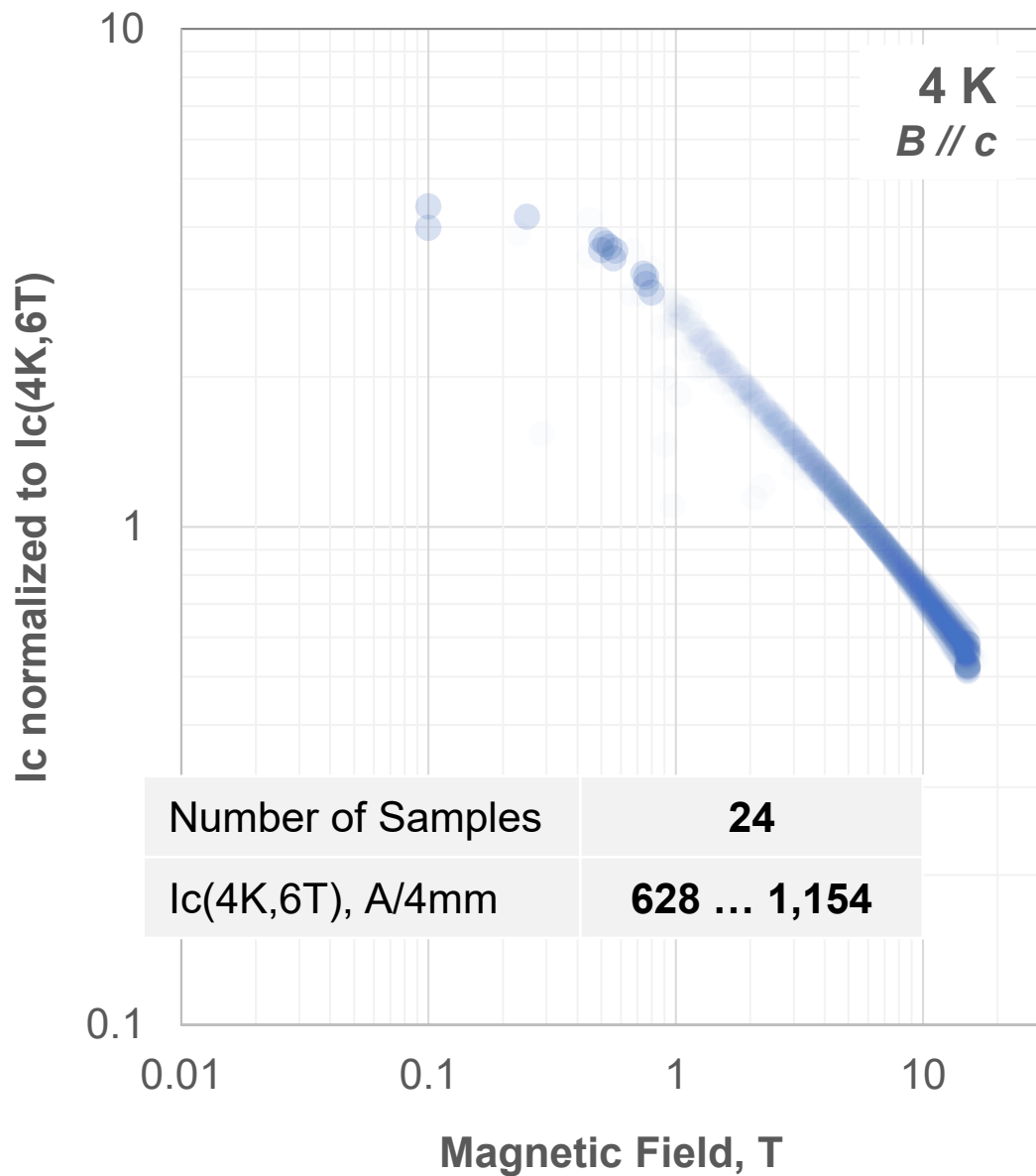
60 – 77.5 K

Instrument: [SuperCurrent, HTS-110](#)

Critical Current

Field dependence @4K

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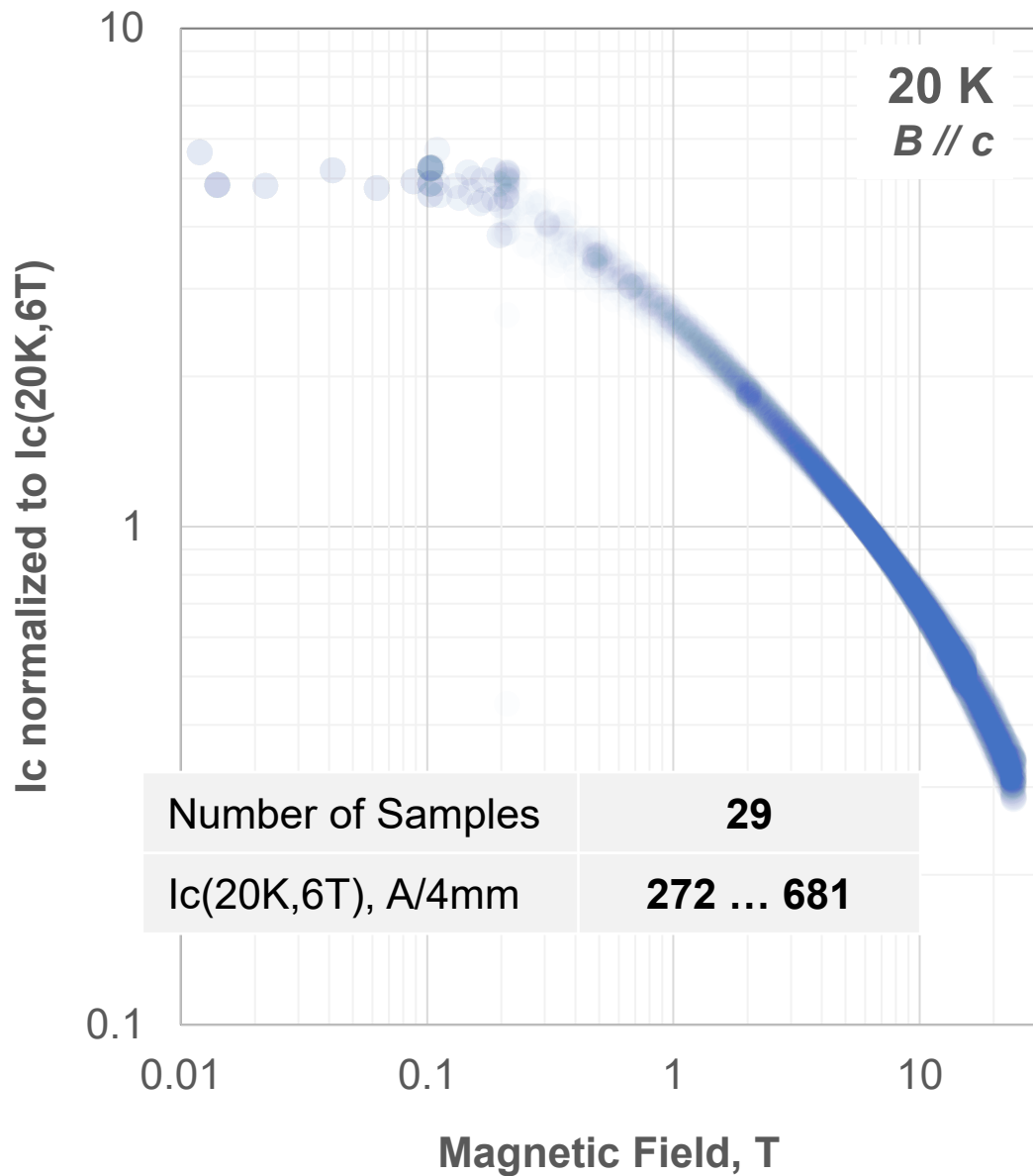


Sources: [High Field Laboratory at Tohoku University
http://doi.org/10.1109/TASC.2025.3543797](http://doi.org/10.1109/TASC.2025.3543797)

Critical Current

Field dependence @20K

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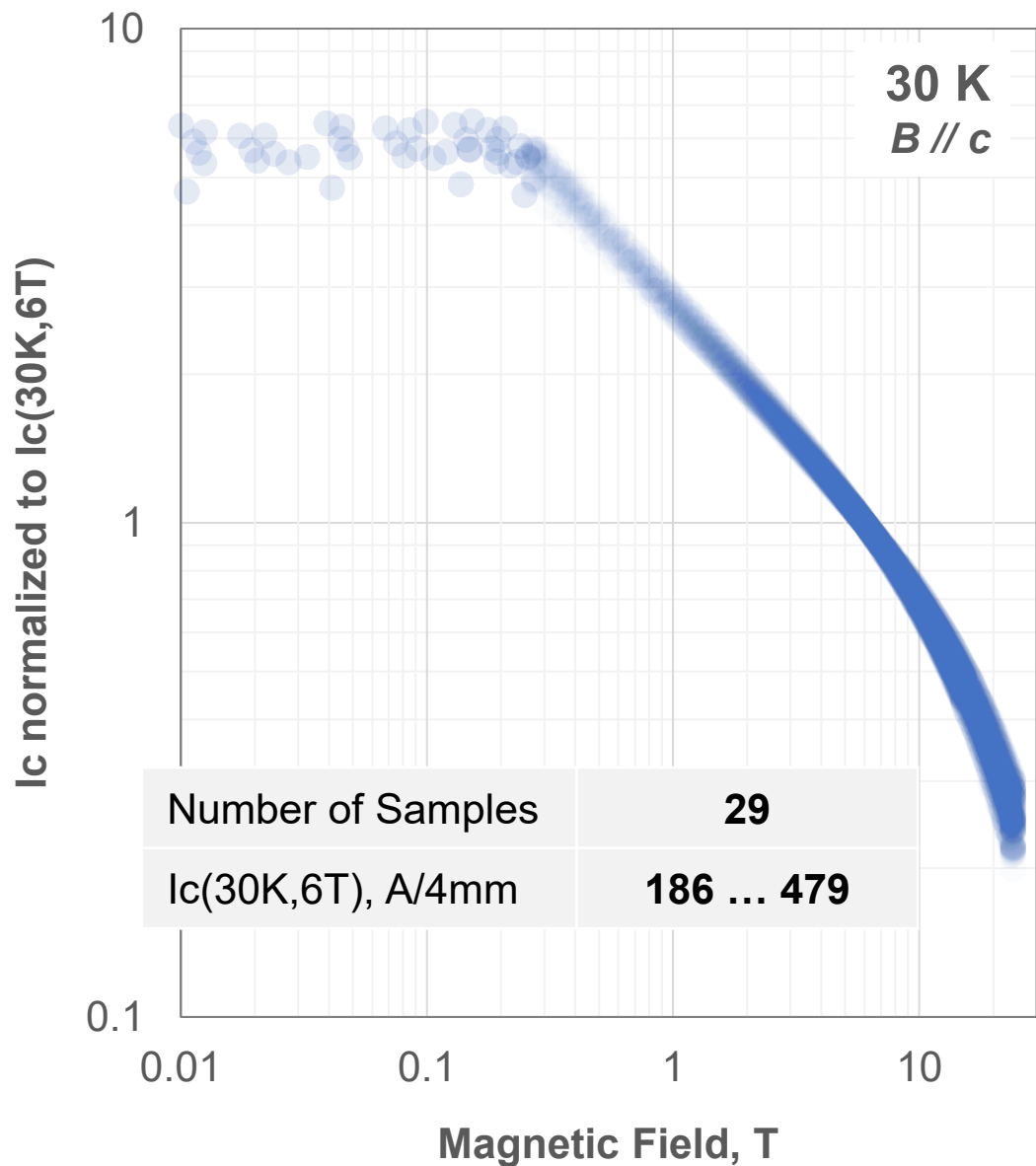


Instrument: [High Field Laboratory at Tohoku University](#)

Critical Current

Field dependence @30K

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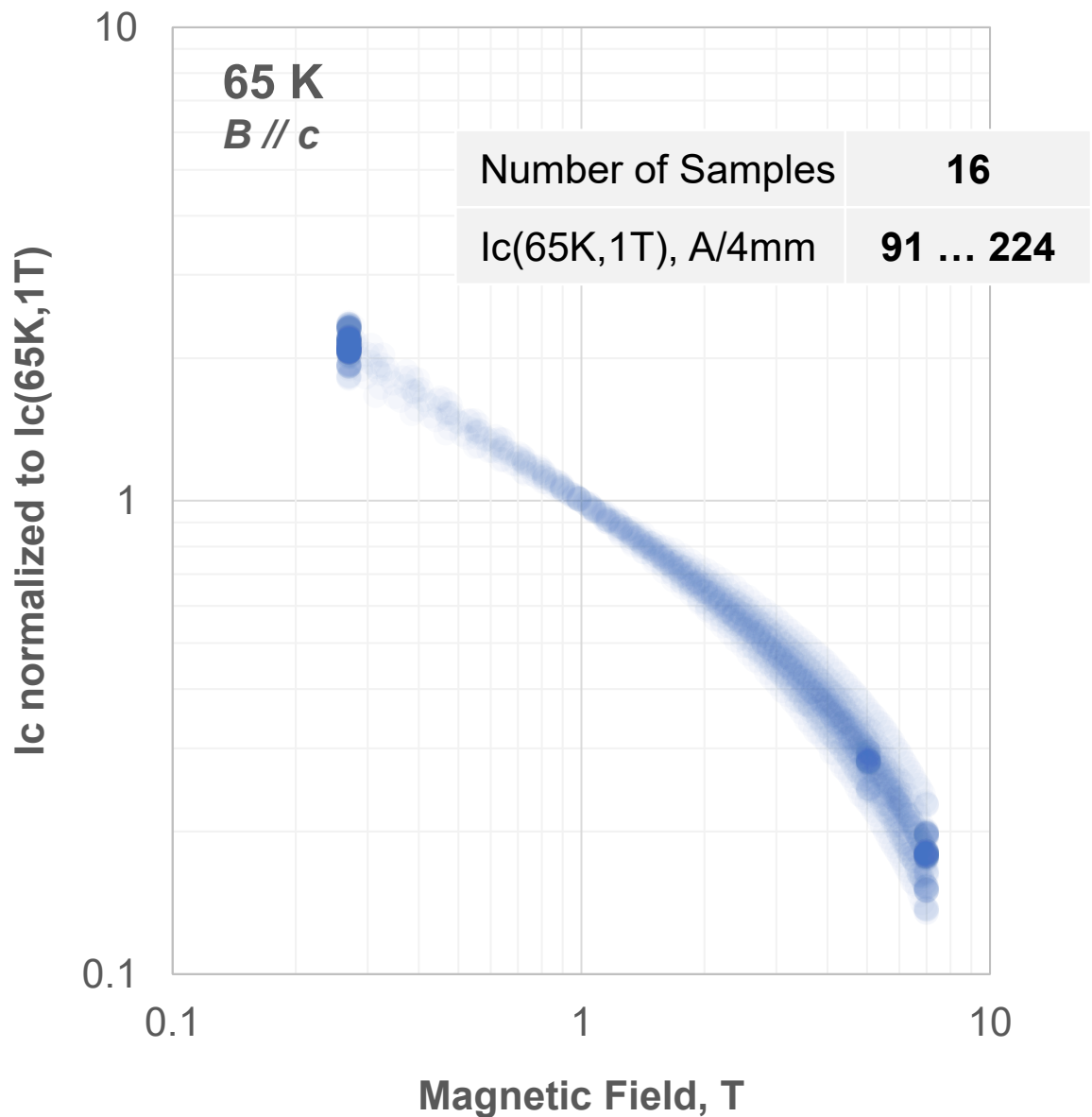


Instrument: [High Field Laboratory at Tohoku University](#)

Critical Current

Field dependence @65K

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Instrument: [High Field Laboratory at Tohoku University](#)

Recommended Lift Factors

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Lift Factor = min Ic (B,T) / min Ic (77K, self field)

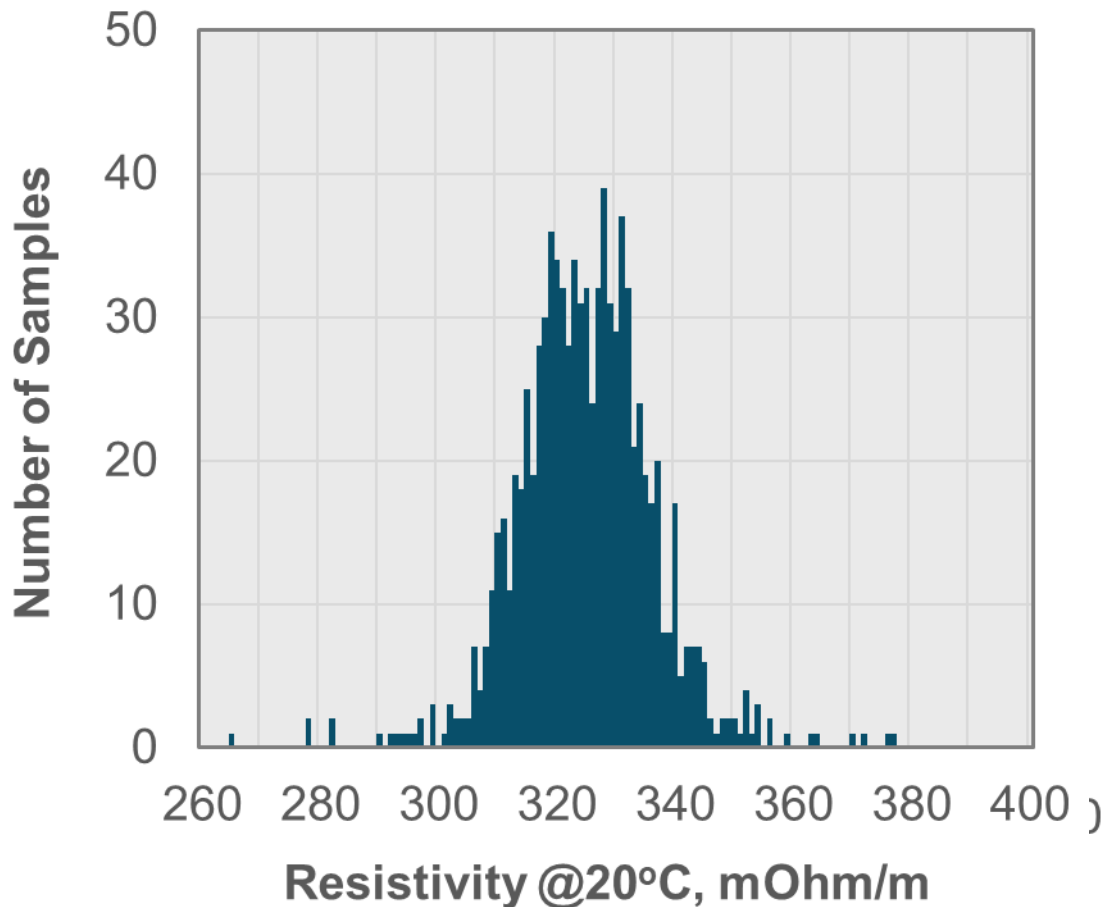
	4.2K	20K	30K	40K	50K	60K	70K	77.3K
0.00 T	15.9	11.7	9.1	6.8	5.0	3.3	1.9	1.00
0.01 T		11.7	9.1	6.8	4.9	3.3	1.9	0.91
0.03 T		11.6	9.0	6.7	4.8	3.2	1.7	0.74
0.1 T		11.3	8.6	6.2	4.3	2.6	1.2	0.46
0.3 T		9.8	7.0	4.7	3.0	1.8	0.76	0.27
0.5 T	11.5	7.3	5.8	3.9	2.5	1.4	0.58	0.22
1 T	9.3	6.1	4.2	2.8	1.8	1.0	0.40	0.12
2 T	7.0	4.3	2.9	2.0	1.3	0.69	0.26	
3 T	5.8	3.5	2.4	1.6	1.0	0.54	0.19	
4 T	5.0	3.0	2.1	1.4	0.88	0.46	0.15	
5 T	4.4	2.6	1.8	1.2	0.76	0.38	0.11	
6 T	4.0	2.3	1.7	1.1	0.68	0.33		
7 T	3.7	2.1	1.5	1.0	0.60	0.28		
8 T	3.4	1.9	1.3	0.89	0.53	0.24		
9 T	3.2	1.8						
10 T	3.0	1.7						
12 T	2.7	1.5						
14 T	2.5	1.4						
16 T	2.3	1.3						
18 T	2.1	1.2						
20 T	2.0	1.1						

Instrument: SuperCurrent for 1-8T, various high field magnets for high field

Electric Properties

Resistivity @20°C

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Number of Samples	886
Average Value, mOhm/m	324
Standard Deviation, mOhm/m	14

Instrument: HP/Keithley, in-house 4-point measurement

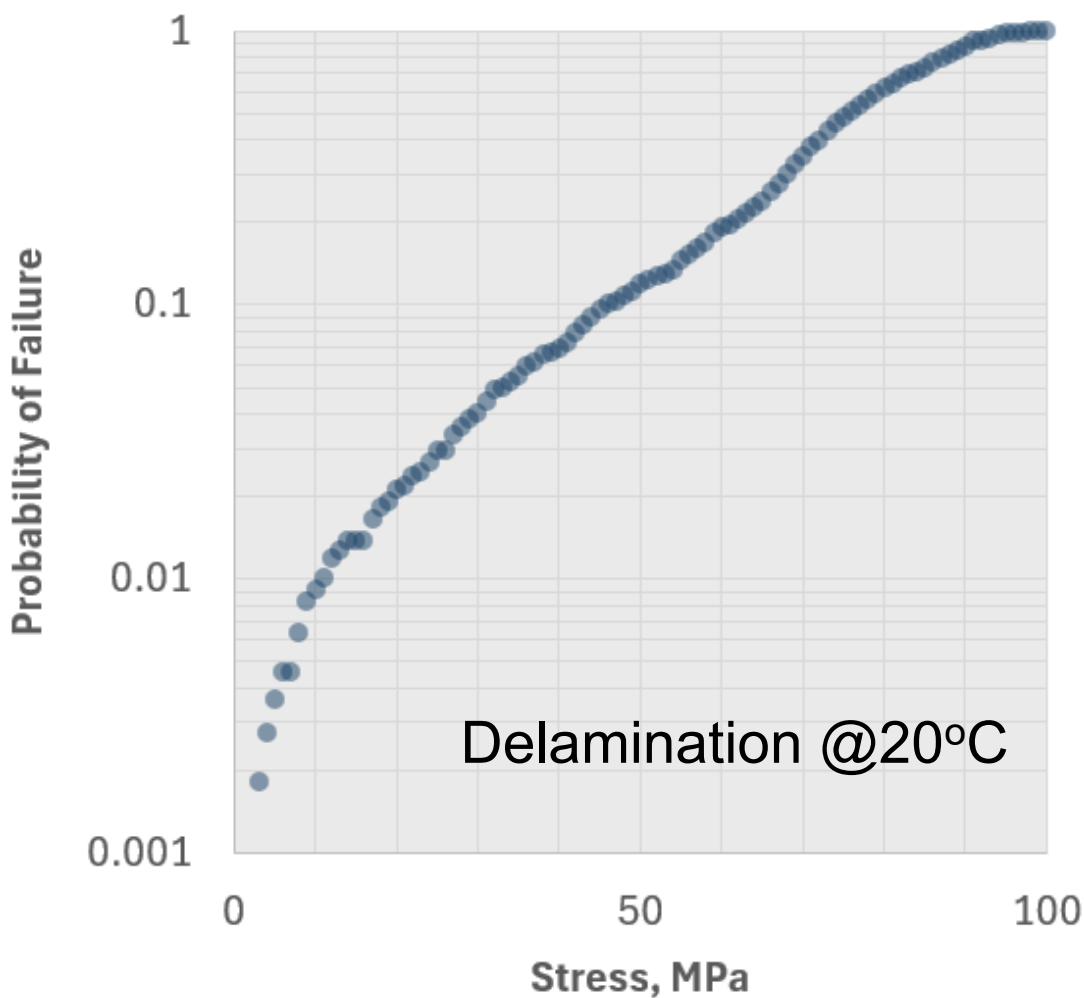
Delamination strength

Double Bending Diameter

Mechanical Properties

Delamination Strength

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Number of Samples

1,322

Average Value, MPa

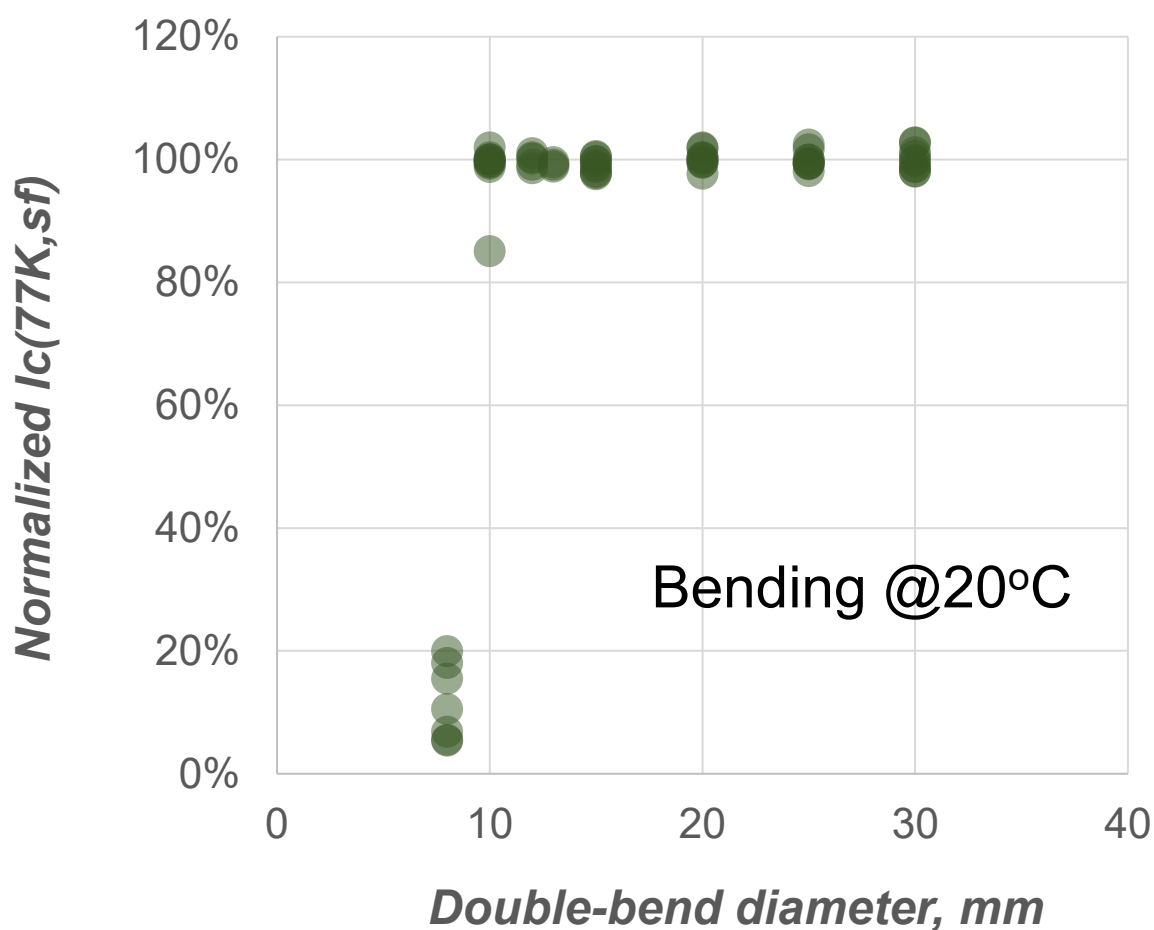
71

Instrument: QUAD, Romulus Universal Mechanical Strength Tester

Mechanical Properties

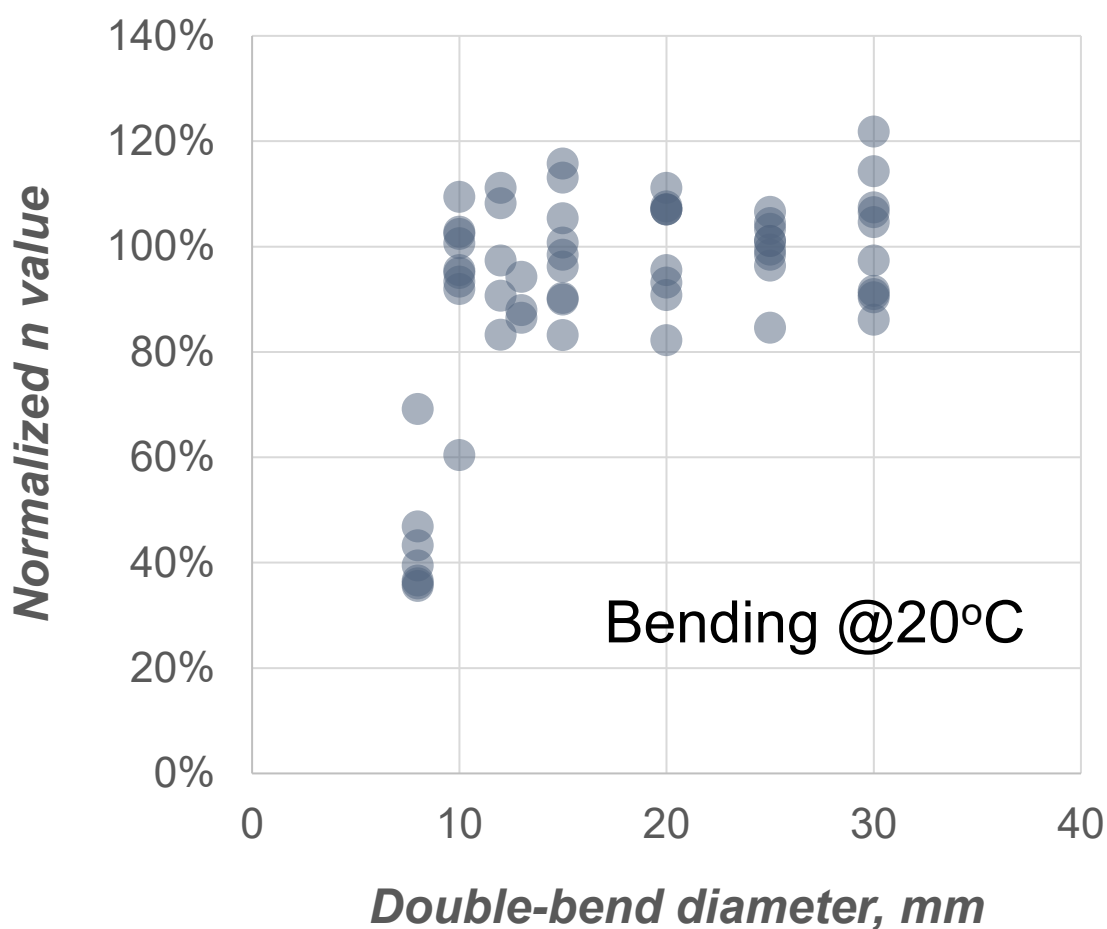
Double Bending Diameter

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Source: in-house measurements

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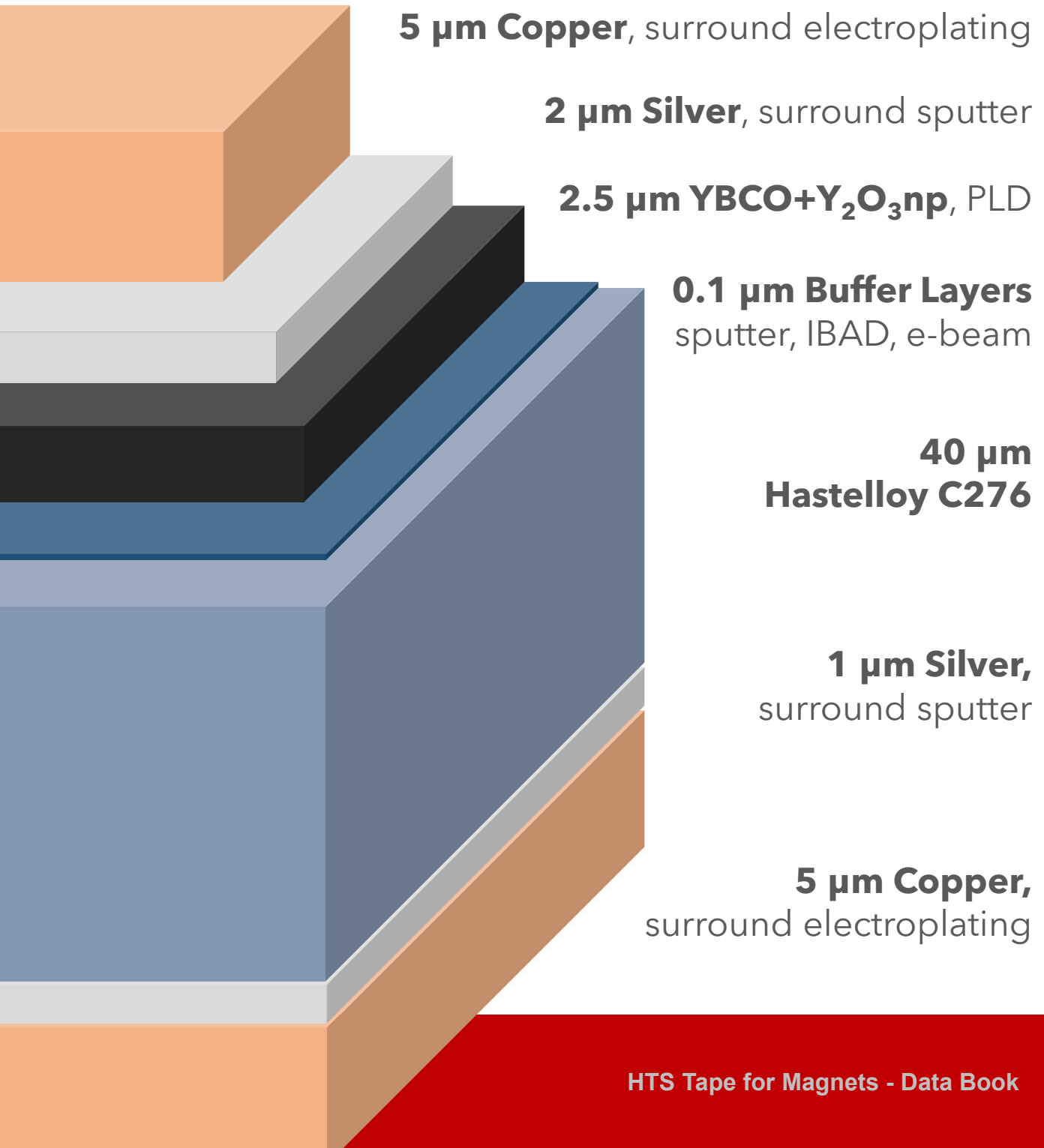
Source: in-house measurements

Resistance of lap joints made in-house with PbSn solder

Length, mm	Width, mm	Ic, A/4mm	R, nOhm	R, nOhm*cm ²
75	4	220	9.4	28
75	4	214	8.1	24
75	4	218	8.4	25
75	4	214	9.0	27
75	4	224	11.2	34
75	4	229	9.6	29
75	4	170	9.7	29
75	4	230	8.1	24
75	4	226	8.3	25
75	4	225	9.3	28
75	4	222	8.9	27
75	4	226	6.6	20
		Average >>>	8.9	26.7

Source: Sorensen/Keithley, in-house 4-point measurement

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5 μm Copper, surround electroplating

2 μm Silver, surround sputter

2.5 μm YBCO+Y₂O₃np, PLD

0.1 μm Buffer Layers
sputter, IBAD, e-beam

40 μm
Hastelloy C276

1 μm Silver,
surround sputter

5 μm Copper,
surround electroplating